

Estimating Transfer Costs from Real Human Mobility Data in Urban Transit

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Abstract

This study proposes a method to quantitatively estimate transfer costs in urban public transportation using large-scale IC card data from Japan's Kansai region. Unlike traditional approaches that rely on surveys, our model empirically infers transfer penalties from observed mobility. We extend the cost-based radiation model developed for inter-city travel to intra-city contexts by incorporating station-level land use. Transfer costs are estimated by calibrating the travel cost function to maximize the Common Part of Commuters (CPC), a commonly used metric for comparing predicted and observed flows. Our findings indicate that the optimal uniform transfer cost corresponds to approximately 30 minutes of generalized travel time. Furthermore, disaggregated analysis reveals significant variation in transfer costs depending on physical and institutional factors, with the highest burdens arising from inter-station, inter-operator transfers. These results provide insights for designing more effective transfer policies.

CCS Concepts

• **Applied computing** → **Transportation**; • **Information systems** → **Geographic information systems**; • **Mathematics of computing** → Maximum likelihood estimation.

Keywords

Human mobility modeling, Radiation model, Transfer cost estimation, Urban transportation, Transfer cost, Public transit optimization, Smart card data, Mobility data analytics

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1 Introduction

1.1 Background

With the rapid growth of urban populations worldwide, more people are moving to metropolitan areas, and cities face an increasing demand for urban transportation. For example, urban passenger transport demand is expected to grow by over 75% from 2019 to 2050[5]. This shift is intensifying urban challenges such as CO_2 emissions and transit congestion, underscoring the need for sustainable and efficient urban mobility solutions. Sustainable urban mobility requires reducing dependency on cars and improving the use of public transport. A critical factor is the mitigation of transfer-related burdens, which substantially impact user's transportation choices. In Japan, particularly in metropolitan areas, multiple private rail operators are densely interconnected, and passengers often transfer between lines of different operators. This institutional fragmentation makes it especially important to consider the heterogeneous costs associated with different transfer types. We define *transfer cost* as transfer-related burdens that discourage their inclusion in route selection. Prior studies have estimated transfer costs via surveys, limited by scope and precision. Recent human mobility data, such as GPS advances, have enhanced data-driven analysis. Leveraging this progress, we employ IC card data to quantitatively analyze the role of transfer costs in shaping urban mobility behavior.

1.2 Purpose

The purpose of this study is to estimate transfer costs in urban public transportation using large-scale train mobility data. First, we demonstrate that the cost-based radiation model, originally designed for inter-city mobility, can be effectively adapted to intra-city travel by incorporating station-level land use characteristics. We then show that the additional travel cost incurred by transfers, referred to as transfer cost, can be quantitatively estimated from actual human mobility patterns.

1.3 Terminology: Travel Cost and Transfer Cost

In this study, we define *travel cost* as the total travel burden. This includes riding time, fare, and transfers. To quantify this burden,

we employ a *travel cost function*, which integrates various factors into a single cost metric. The concept of *generalized time* plays a central role in unifying different aspects of travel experience on a single scale.

A key component of the travel cost is the *transfer cost*, which specifically refers to the disutility experienced when changing lines or operators. Transfer cost often reflects walking distance, waiting time, fare complexity, and psychological discomfort. For comparability, we express this value in terms of *generalized time*, which represents all these burdens as equivalent riding time.

2 Related Research

2.1 Human Flow Analysis

A wide range of studies have used human mobility data for urban analysis. Among them, research that uncovers human behavioral patterns plays a key role in urban planning and transportation policy design. For example, Faroqi attempted to infer trip purposes from smart card data. [3] Other studies have analyzed behavioral changes in response to transportation infrastructure updates. For example, investigate the impact of newly opened rail lines using IC card data.[7]

Despite such advances, few studies have attempted to quantitatively estimate travel costs based on observed human mobility data.

2.2 Human Flow Prediction Models and Metrics

2.2.1 Human Flow Prediction Models. Ma approaches have attempted to model inter-city flows based on geographical distance and city size. The most traditional one is the gravity model, which assumes that the number of travelers between two locations is proportional to their population and inversely related to the distance between them.[2] Simini introduced the radiation model, which addresses several limitations of the gravity model and has gained attention due to its parameter-free formulation and universality.[12] However, the radiation model has mainly been applied to inter-city flows such as between states or municipalities and has seen limited application at the intra-city, station-level scale.[10] An extension known as the cost-based radiation model has also been proposed. Unlike the original model, which uses distance as the determinant of costs, this model allows for flexible definitions of travel costs.[9] Additionally, machine learning-based mobility models have been introduced, but they often suffer from high black-box complexity. While high prediction accuracy, they are less suitable for interpretability and policy analysis.[11]

2.2.2 Evaluation Metrics. To evaluate the performance of the proposed mobility model, ma indicators were proposed such as Absolute Error(AE), Relative Error(RE), and Common Part of Commuters (CPC).

AE measures the total deviation between predicted and observed flows. It is defined as $AE_{ij} = |T_{ij} - T_{ij}^*|$ where T_{ij} and T_{ij}^* denote the predicted and actual number of travelers between zones i and j , respectively. RE normalizes the deviation by the actual flow volume, offering a scale-independent perspective $RE_{ij} = \frac{|T_{ij} - T_{ij}^*|}{T_{ij}^*}$. CPC measures the overlap between predicted and observed flows and

is defined as: $CPC = \frac{2 \sum_{ij} \min(T_{ij}, T_{ij}^*)}{\sum_{ij} T_{ij} + \sum_{ij} T_{ij}^*}$. CPC takes values between 0 and 1, where a value of 1 indicates perfect agreement. A notable strength of CPC is its emphasis on correctly predicting large flows.

2.3 Travel Cost Analysis

Traditional models rely on survey data, which are limited in coverage, scale, and behavioral accuracy, making them difficult for detailed urban mobility analysis. In contrast, recent studies on human mobility increasingly leverage large-scale behavioral data such as smart card records. Following this trend, our study estimates transfer costs using actual mobility trajectories within a unified travel cost framework.

3 Data

3.1 Railroad Human Flow Data

3.1.1 Data Abstract. This study utilizes IC card data collected from public transportation systems affiliated with the Surutto KANSAI Association in the Kansai region of Japan. The association includes 61 railway, subways and bus operators, covering most transit networks in Kansai, Japan's second-largest metropolitan area. Data include user IDs, timestamps of boarding and alighting, and the corresponding station and line names.

To calculate travel costs, information on inter-station *travel distance*, *travel time*, *transfer details*, and *fare* was collected. The transfer details include the alighting station, alighting line, boarding station, and boarding line. A total of 137,107 inter-station travel records were extracted.

3.1.2 Preprocess. To avoid irregular effects caused by the COVID-19 pandemic, we use mobility data in 2019. In addition, different travel purposes imply varying perceptions of transfer costs, and mixing them would confound estimation due to heterogeneous decision criteria. To isolate behavior that more consistently reflects discretionary mobility patterns, we focus only on leisure-related trips. Leisure-related travel allows people to choose their destinations with a certain degree of freedom, making it well-suited to the aim of this study. Consequently, we exclude trips that likely represent routine commuting activity. Specifically, we removed a trip that satisfied at least one of the following conditions: (1) the same user used identical boarding and alighting stations more than 10 times within a month; (2) the trip was made using a commuter pass; (3) the trip occurred on a weekday. After that process, the number of trips per day was 2,832,362.

3.2 Station Activity Data

In human flow modeling, the population of a state or municipality is typically used to represent the size of a city. However, since this study models mobility at the station level, population data cannot be directly applied. We use the number of restaurants within a 300-meter radius of each station as a proxy for station scale instead. As noted in Section 3.1.2, this study focuses on leisure-related trips, and the number of nearby restaurants is considered a suitable indicator due to its strong association with leisure activities. The data were obtained by the "Google Cloud Places Aggregate API".

4 Proposed Method

4.1 Cost-based Radiation Model for Intra-city Application

Our proposed approach extends the applicability of the radiation model, originally designed for intercity travel to intra-city mobility. The original cost-based radiation model is defined as follows:

$$T_{ij}(n_i, n_j, s_{ij}) = \frac{n_{all}}{n_{all} - n_i} T_i \frac{n_i n_j}{(n_i + s_{ij})(n_i + n_j + s_{ij})} \quad (1)$$

T_{ij} is the number of travelers from origin i to destination j , T_i is the number of travelers from origin i to the all destinations, n_i and n_j represent the scale of each city (e.g., population), s_{ij} is the total scale of all locations with lower travel cost than from i to j , and n_{all} is the sum over all n_i . In this model, the travel cost function affects s_{ij} , which represents the cumulative scale of competing destinations that are more accessible than the destination j from the origin i . Therefore, the travel cost function determines the effective spatial reach of travelers.

While the model works well at coarse geographical levels, it fails to capture the functional heterogeneity of stations at finer granularity. To solve this issue, we focus exclusively on flows from residential to commercial areas, extracting only relevant trips for modeling. We cluster stations using boarding and alighting patterns (aggregated every two hours and separated by weekdays and holidays), following existing methodologies.[1, 6] The resulting clusters clearly distinguish residential stations from commercial stations.

4.2 Estimating Transfer Cost

As the travel cost function differs, the predicted results and CPC are affected. We assume that the most plausible transfer cost corresponds to the value that maximizes CPC.

We first consider a simplified model in which a single uniform transfer cost is assumed for all transfers in eq(2).

$$\text{Cost} = (\text{Travel Time}) + p \times (\text{Num Transfer}) \quad (2)$$

Here, p is the transfer cost for one transfer. NumTransfer denotes the number of transfers of that type in one trip. For example, when comparing two travel cost functions—(a) $\text{Travel Time} + 10 \times \text{Num Transfers}$, and (b) $\text{Travel Time} + 30 \times \text{Num Transfers}$ if (a) yields a higher CPC, it suggests that the actual transfer cost is closer to 10 minutes rather than 30. To avoid multicollinearity, fare which was strongly correlated with distance was excluded from the travel cost function. In addition, we introduced multiple transfer cost parameters to account for heterogeneity. Transfer costs are influenced by both physical infrastructure and institutional boundaries. To capture this heterogeneity, we extend our model by assigning distinct transfer cost parameters to different transfer types. Transfers are categorized based on station and operator continuity, resulting in four types: (1) Inter-station, Inter-operator, (2) Intra-station, Inter-operator, (3) Intra-station, Intra-operator, (4) Inter-station, Intra-operator. Each type is associated with a separate transfer cost parameter in the travel cost function.

$$\text{Cost} = (\text{Travel Time}) + \sum_{i \in \text{Transfer Types}} p_i \times (\text{Num Transfer}_i) \quad (3)$$

Here, p_i is the additional transfer cost for transfer type i , and num_transfer_i denotes the number of transfers of that type in one trip. In our implementation, p_i is treated as an integer parameter in the range of 0 to 100, and its optimal value is determined using the Tree-structured Parzen Estimator (TPE) algorithm provided by Optuna.

5 Results

5.1 Cost-based Radiation Model for Intra-city Application

Figure 1 illustrates the results of our proposed method, where the model is restricted to flows from residential to commercial areas. CPC calculated in the same origin and destination pairs increased from 0.465 without cluster information to 0.529 with cluster information. Even with the gravity model, the CPC increased from 0.513 to 0.544 with the same method. The gravity model achieves higher accuracy than the radiation model, as its parameters are tuned based on actual data.

These results suggest that intra-city mobility involves structural flow patterns that cannot be fully captured by traditional models relying solely on distance or population. Explicitly modeling functional urban characteristics, such as land-use-based station types, is important for improving prediction accuracy.

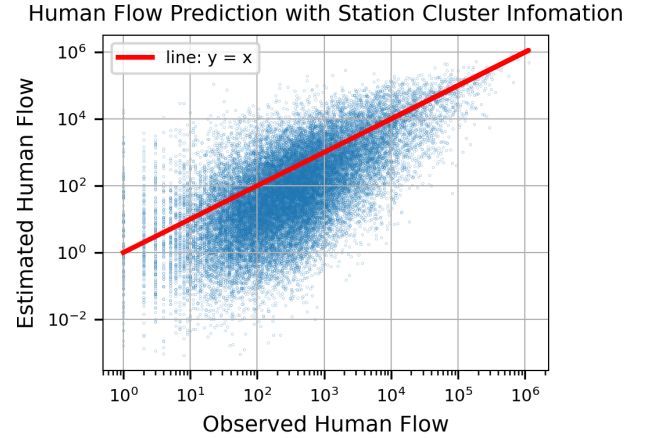


Figure 1: Comparison between observed and estimated human flows using a station-cluster-enhanced radiation model. Each dot represents an OD pair. The red line denotes the ideal line $y = x$. Incorporating station cluster information into the model improves alignment with real mobility patterns.

5.2 Estimation of Transfer Cost

5.2.1 Estimation with a Single Transfer Cost. Figure 2 shows the relationship between the transfer cost parameter p and CPC. The blue line shows the CPC trend when using the all dataset, while the light blue lines represent the CPC trends for each user group constructed by dividing users into four segments. The peak is observed at $p = 30.0$, indicating $p = 30$ is the optimal value.

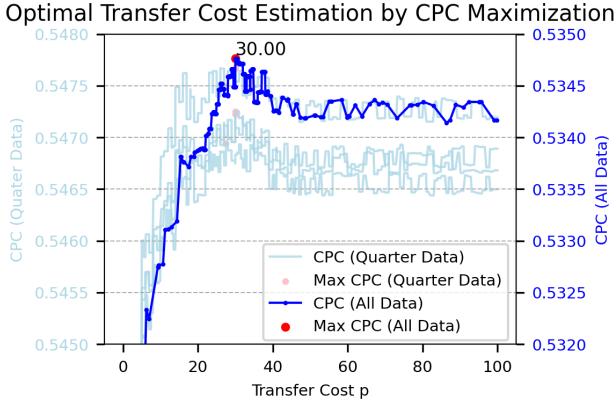


Figure 2: Relationship between transfer cost parameter p and CPC. This convex shape implies the existence of an optimum, suggesting that human route choice behavior reacts systematically to changes in perceived transfer cost.

Three key observations emerge from this result: First, the CPC behaves as a smooth and convex function for the transfer cost parameter p , indicating the presence of an optimal cost level. Second, the optimal unified transfer cost is found to be approximately 30 minutes, which is slightly longer generalized travel time reported in previous studies. Third, the consistency of the CPC curve across user groups reinforces the robustness of the estimation. All four user groups exhibit similar CPC curves with peaks $p = 30$ (30.0, 30.0, 30.0, and 27.5 for each group), suggesting that the optimal transfer cost is robust.

5.2.2 Estimation of Transfer Costs by Transfer Type. The optimal combination that maximizes CPC is summarized as follows.

- Inter-station, Inter-operator: 87 min
- Intra-station, Inter-operator: 70 min
- Inter-station, Intra-operator: 73 min
- Intra-station, Intra-operator: 31 min

The results are consistent with intuition: transfers between different stations and different operators incur the highest transfer cost due to increased walking distance and fare complexity.

6 Discussion

6.1 Validity of the Estimated Transfer Cost

In the experiment, the estimated transfer cost was approximately 30 minutes. However, prior survey-based studies report that a generalized transfer cost is around 15 minutes, lower than our estimate, even considering the effect of the fare [4]. Several factors may explain this discrepancy. The first factor is the differences in transfer context. The previous study[4] measured transfer cost under the assumption of traveling to a predetermined destination. In contrast, our study estimates transfer costs in the context of destination choice. In such scenarios, transfer may exert greater influence.

The second one is the difference between behavioral data and stated preference. Our estimation is based on actual IC card records, reflecting revealed behavior. Survey-based studies rely on stated

preferences, which are prone to cognitive bias and hypothetical assumptions, potentially underestimating actual behavioral transfer costs.

6.2 Impact of Transfer Types

We revealed substantial variation in transfer costs. Specifically, transfers involving different stations and different operators incurred the highest costs (87 minutes), whereas transfers within the same station and operator were the least costly (30 minutes). This suggests that institutional boundaries significantly influence transfer costs. These findings underscore the importance of context-sensitive modeling for urban transport planning. Improvements such as streamlined intra-operator transfer pathways and unified fare systems could reduce transfer costs.

Although an 87-minute transfer cost may appear unusually high, it may align with travel behaviour in the Kansai region. Because the Kansai region’s private railway operators have deliberately cultivated both residential suburbs and bustling commercial hubs along each line, most daily and leisure activities can be satisfied within a single corridor [8]. Consequently, passengers rarely need to transfer, and this fact engenders an aversion to transferring—namely, a heightened perceived transfer cost. Note that estimated transfer cost does not represent the actual time spent transferring, but rather reflects the psychological and behavioral deterrence associated with making a transfer, as inferred from observed route choices.

7 Conclusion and Future Work

This study proposed an extension of the cost-based radiation model to intra-city mobility by incorporating station-level land use. The model improved prediction accuracy and enabled data-driven estimation of transfer costs. Furthermore, the analysis provided empirical insights into how operator fragmentation impacts human mobility, revealing quantifiable burdens associated with inter-operator transfers. These findings may inform future transportation network design and policy interventions.

Nonetheless, our study has several limitations. First, it does not fully account for alternative modes, such as buses or parallel rail lines, which may lead to an underestimation of actual passenger flows. Second, the cost-based radiation model exhibits discontinuous behavior concerning transfer cost parameters, making it difficult to use gradient-based optimization techniques and poses computational challenges. Moreover, the parameter space grows exponentially with the number of transfer pattern combinations, underscoring the need for future research on model simplification or structure-aware optimization strategies. Finally, this method has limited generalizability. Although this method can be applied to other regions, differences in urban environments affect the interpretation of results. Additionally, for other than leisure trips like commuting, where destinations are fixed, transfer costs may have little influence on destination decision.

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